

Railway DPMS Application - Project Brief

Overview

The Railway DPMS (Department Performance Management System) is a comprehensive web-based application developed using Python Flask framework, designed to streamline the management and tracking of Key Performance Indicators (KPIs) across various departments within a railway organization. The system implements a sophisticated multi-level approval workflow with role-based access control, ensuring accountability, transparency, and efficiency in performance tracking processes. By digitizing the entire KPI lifecycle from data entry to final approval, the application eliminates manual paperwork, reduces processing time, and provides a centralized platform for all stakeholders to monitor and evaluate departmental performance.

Backend Infrastructure

The application is built on Flask, a lightweight and flexible Python web framework, which provides robust routing capabilities, session management, and template rendering functionalities. The database layer utilizes SQLAlchemy ORM with connection pooling to ensure stable database connectivity, featuring pre-ping mechanisms to prevent connection timeouts and session recycling for optimal performance. File handling is implemented through Werkzeug's secure filename utilities, with strict validation for PDF uploads limited to 10MB maximum size.

Frontend Implementation

The user interface employs Jinja2 templating engine for dynamic HTML rendering, with role-specific dashboards that adapt to different user permissions. The application features responsive table layouts, bulk action capabilities, and intuitive navigation flows tailored to each user role's specific requirements.

Role-Based Workflow and User Hierarchy

The system implements a comprehensive five-level approval hierarchy, each with distinct responsibilities and access permissions:

Level 1 - Department Employees are responsible for entering monthly KPI data, saving drafts for later completion, and submitting finalized data for approval. They can view returned items with remarks and make necessary corrections based on feedback from superiors. This level serves as the foundation of the data entry process, ensuring all performance metrics are accurately recorded at the source.

Level 2 - Head of Department (HOD) reviews all submitted KPIs, generates PDF approval templates for physical signatures, uploads signed PDF documents, and either approves or returns KPIs with detailed remarks. The HOD plays a critical role in validating data accuracy before it proceeds to higher authorities, with the ability to perform both single and bulk approvals for efficiency.

Level 3 - Nodal Officer reviews approved KPIs from all departments, verifies document completeness, and forwards selected KPIs to the Additional Divisional Railway Manager (ADRM) for further processing. This role acts as a central coordination point, ensuring all departmental data meets organizational standards before escalation.

Level 4 - ADRM examines forwarded KPIs, performs quality checks, and either forwards them to the Divisional Railway Manager (DRM) or returns them to the Nodal Officer with constructive feedback. This level adds an additional layer of scrutiny to maintain data integrity.

Level 5 - Divisional Railway Manager (DRM) serves as the final approval authority, reviewing all forwarded KPIs and granting final approval. Upon DRM approval, the data is permanently frozen and copied to the *approved_data table*, establishing an immutable historical record. The DRM can also return KPIs to ADRM for revision if discrepancies are identified.

Review and Approval Chain

The HOD reviews submitted KPIs, generates PDF templates for signing, uploads signed documents, and approves the records, which then move to the Nodal Officer's dashboard. The Nodal Officer performs a secondary review and forwards approved KPIs to ADRM, who conducts another level of scrutiny before forwarding to DRM for final

approval. Each approval stage includes the ability to return records with remarks for correction, maintaining a feedback loop that ensures data accuracy.

Finalization and Archiving

Upon DRM approval, the data is permanently frozen and copied to the approved_data table, creating an immutable historical record. This archiving mechanism ensures that approved performance data remains accessible for future reference while preventing unauthorized modifications, establishing a reliable audit trail for organizational performance tracking.

```
C:\Users\akhils\OneDrive\Desktop\railway3>tree /F
Folder PATH listing for volume Windows-SSD
Volume serial number is A205-7051
C:..
|  app.py
|  config.py
|  create_user.py
|
|  ---models
|      user.py
|
|  ---routes
|      auth.py
|      _init_.py
|
|  ---static
|      |
|      |  ---images
|      |      Emblem_of_India_with_transparent_background.png
|      |      images.png
|      |      railway_bg.jpg
|
|  ---templates
|      adrm.html
|      department_form.html
|      drm.html
|      hod_review.html
|      login.html
|      nodal.html
|      return_form.html
|
|  ---uploads
|      |
|      |  ---hod_approvals
|      |      |
|      |      |  ---signed_pdfs
|      |
|  ---__pycache__
```

app.py

This is the **main application file** that contains all the routing logic, database configuration, and business logic. It's the central controller that:

- Database initialization (SQLAlchemy)
- Route definitions for all user roles
- Session management
- File upload/download handling
- PDF generation with ReportLab
- Bulk operations logic
- Authentication & authorization

config.py

This file stores all **environment-specific configurations** for the application.

models/user.py

This defines the **User database schema** and interactions.

routes/**auth.py**

This file would contain all **authentication-related routes**

Static folder contain image files

templates/

File	Purpose
login.html	User login page
department_form.html	LEVEL1: KPI data entry with drafts/submit
hod_review.html	LEVEL2: Review, approve, return, upload documents
nodal.html	LEVEL3: Forward approved KPIs to ADRM
adrm.html	LEVEL4: Forward to DRM or return to employee
drm.html	LEVEL5: Final approval and freezing data
return_form.html	Rejection form with remarks input

uploads/

1)hod_approvals/ -

Contains **all signed PDF documents** and their metadata:

Document Files

Files follow naming conventions:

{timestamp}_{kpi_id}_{original_name}.pdf

Example:

20260620_211944_HOD_Approval_Template_b_Ordinary_Working_Expenses_Net_Spending_Limits_JUNE_2026.pdf

bulk_*.pdf - Bulk Upload Files

2)signed_pdfs/ - Signed Documents

DB schema

1) departments

Purpose: Master table storing all departmental information within the railway organization.

Column	Type	Description
id	integer	Primary key, auto-generated unique identifier for each department
dept_code	character varying	Unique department code (e.g., "FIN", "HR", "OPS") for quick reference
dept_name	character varying	Full department name displayed throughout the application

Key Relationships: Referenced by users, kpis, and workflow tables as foreign key

2) users

Purpose: Manages all system users with authentication credentials and role assignments.

Column	Type	Description
id	integer	Primary key, auto-generated unique identifier for each user
username	character varying	Unique login identifier for authentication
password_hash	text	Securely hashed password for user authentication
role	character varying	User role: LEVEL1, LEVEL2, LEVEL3, LEVEL4, LEVEL5, or ADMIN
department_id	integer	Foreign key linking user to their department
created_at	timestamp	Auto-generated timestamp of user account creation

Key Relationships: References departments(id); Referenced by monthly_data, approved_data, and audit_logs

Purpose: Implements role-based access control and departmental data ownership, ensuring users access only authorized data and perform only permitted actions.

3)kpis

Purpose: Defines all Key Performance Indicators tracked by the system.

Column	Type	Description
id	integer	Primary key, auto-generated unique identifier for each KPI
section_name	text	Section/category under which the KPI falls
kpi_name	text	Descriptive name of the performance indicator
unit	character varying	Measurement unit (e.g., %, Rs., Count, MT)
department_id	integer	Foreign key linking KPI to its owning department
display_order	integer	Controls UI sorting and presentation order (default: 0)
is_auto_fetch	boolean	Flag for automatic data fetching from external sources
is_active	boolean	Enables/disables KPI for data entry (default: true)
department_name	text	Denormalized department name for quick display
data_source	text	Indicates external data source if auto-fetch is enabled
row_type	character varying	Classification for data presentation
data_from	character varying	Source system or origin of data
serial_no	character varying	Reference number for external system integration

Key Relationships: References departments(id); Referenced by monthly_data, approved_data, and annualtarget_info

Purpose: Serves as the master reference for all performance metrics, enabling flexible KPI management and department-specific tracking.

3)monthly_data

Purpose: Core transactional table storing all KPI performance submissions throughout the workflow.

Column	Type	Description
id	integer	Primary key, auto-generated unique identifier for each record
kpi_id	integer	Foreign key referencing the KPI being measured

month	character varying	Month of performance data (e.g., "JANUARY", "FEBRUARY")
year	integer	Year of performance data
performance_month	numeric	Monthly performance value for the KPI
cumulative_performance	numeric	Year-to-date cumulative performance value
entered_by	integer	Foreign key referencing user who entered the data
created_at	timestamp	Auto-generated timestamp of record creation
status	character varying	Workflow status: DRAFT, SUBMITTED, APPROVED, FORWARDED_TO_ADRM, FORWARDED_TO_DRM, FROZEN, RETURNED
previous_year_value	bigint	Performance value from same month of previous year
cumulative_performance_of_prev_year	bigint	Cumulative performance from previous year
remarks	text	Comments, feedback, or rejection reasons

Key Relationships: References kpis(id) and users(id); Data copied to approved_data upon final approval

Purpose: Manages the complete lifecycle of KPI data, tracking performance metrics through the multi-level approval workflow from draft to final freezing.

4)approved_data

Purpose: Archives final approved performance data after DRM approval.

Column	Type	Description
id	integer	Primary key, auto-generated unique identifier
kpi_id	integer	Foreign key referencing the KPI
month	character varying	Month of approved performance data
year	integer	Year of approved performance data
performance_month	numeric	Approved monthly performance value
cumulative_performance	numeric	Approved cumulative performance value
entered_by	integer	Foreign key referencing user who entered the data
created_at	timestamp	Auto-generated timestamp of approval
status	character varying	Always 'APPROVED' or 'FROZEN' for archived records
previous_year_value	bigint	Previous year comparison value
cumulative_performance_of_prev_year	bigint	Previous year cumulative comparison
remarks	text	Approval remarks or notes
mark	bigint	Additional marking or scoring field

Key Relationships: References kpis(id) and users(id); Populated from monthly_data upon DRM approval

Purpose: Creates an immutable historical record of approved performance data, preventing modifications and maintaining audit integrity.

5)annualtarget_info

Purpose: Stores year-specific annual targets for each KPI.

Column	Type	Description
year	bigint	Calendar year for which the target is set (Primary Key part)
ref_id	integer	Foreign key referencing the KPI (Primary Key part)
annual_target	double precision	Numeric target value for the year (default: 0)

Key Relationships: References kpis(id); Composite primary key (year, ref_id)

Purpose: Enables year-over-year target comparison and performance assessment against organizational goals.